

PROJECT PHOENIX ? MODEL-DRIVEN CALCULATION DOSSIER
CONCEPT CALCULATIONS - NOT FOR SUBMISSION OR EXECUTION

Project: HBM-2026-001
Model: HBM-GEO-2026-001
Model fingerprint: 61c1c56f500192557d603552561dd408cfed09fe704247b26f18343d24f1af0a
Calculation records: 32
Quality checks: 18/18

AREA VOLUME

CAL-A01 ? Extension floor area
Method: width * length
Result: 70.0 m2
References: model=BLD-EXTENSION; drawings=A-101,A-102; reports=R-101; REQ=REQ-102
Status: SIMULATION_ONLY / PROFESSIONAL ACCEPTANCE CRITERION NOT APPLIED

CAL-A02 ? Extension gross floor area
Method: floor_area * storeys
Result: 140.0 m2
References: model=BLD-EXTENSION,L00,L01; drawings=A-101,A-102; reports=R-001,R-101; REQ=REQ-102
Status: SIMULATION_ONLY / PROFESSIONAL ACCEPTANCE CRITERION NOT APPLIED

CAL-A03 ? Extension enclosed volume
Method: floor_area * storey_height * storeys
Result: 448.0 m3
References: model=BLD-EXTENSION; drawings=A-201,A-301; reports=R-101; REQ=REQ-102
Status: SIMULATION_ONLY / PROFESSIONAL ACCEPTANCE CRITERION NOT APPLIED

CAL-A04 ? Existing gross floor area
Method: existing_width * existing_length * storeys
Result: 336.0 m2
References: model=BLD-EXISTING; drawings=A-101,A-102; reports=R-101; REQ=REQ-102
Status: SIMULATION_ONLY / PROFESSIONAL ACCEPTANCE CRITERION NOT APPLIED

CAL-A05 ? Total gross floor area
Method: existing_gross_area + extension_gross_area
Result: 476.0 m2
References: model=BLD-EXISTING,BLD-EXTENSION; drawings=A-101,A-102; reports=R-001; REQ=REQ-102
Status: SIMULATION_ONLY / PROFESSIONAL ACCEPTANCE CRITERION NOT APPLIED

CAL-A06 ? Modelled net space area
Method: sum(space.area_m2)
Result: 388.32 m2
References: model=SP-L00-01,SP-L00-02,SP-L00-03,SP-L00-04,SP-L00-05,SP-L00-06,SP-L01-01,SP-L01-02,SP-L01-03,SP-L01-04,SP-L01-05,SP-L01-06
Status: SIMULATION_ONLY / PROFESSIONAL ACCEPTANCE CRITERION NOT APPLIED

STRUCTURAL LOADS

CAL-S01 ? Service floor load per storey

Method: $\text{floor_area} * \text{floor_area_load}$

Result: 490.0 kN

References: model=BLD-EXTENSION; drawings=S-201; reports=R-201; REQ=REQ-103

Status: SIMULATION_ONLY / PROFESSIONAL ACCEPTANCE CRITERION NOT APPLIED

CAL-S02 ? Service roof load

Method: $\text{roof_area} * \text{roof_area_load}$

Result: 245.0 kN

References: model=BLD-EXTENSION,LRF; drawings=A-401,S-201; reports=R-201; REQ=REQ-103

Status: SIMULATION_ONLY / PROFESSIONAL ACCEPTANCE CRITERION NOT APPLIED

CAL-S03 ? Total service gravity load

Method: $\text{floor_area} * \text{floor_load} * \text{storeys} + \text{roof_area} * \text{roof_load} + \text{wall_allowance}$

Result: 1335.0 kN

References: model=BLD-EXTENSION; drawings=S-201; reports=R-201; REQ=REQ-103

Status: SIMULATION_ONLY / PROFESSIONAL ACCEPTANCE CRITERION NOT APPLIED

LOAD PATH

CAL-S04 ? Average column reaction

Method: $\text{total_service_load} / \text{column_count}$

Result: 148.33 kN

References: model=CONN-001,CONN-002,CONN-003,CONN-004; drawings=S-201; reports=R-201; REQ=REQ-103

Status: SIMULATION_ONLY / PROFESSIONAL ACCEPTANCE CRITERION NOT APPLIED

CAL-S05 ? Representative beam line load

Method: $\text{floor_area_load} * \text{tributary_width}$

Result: 24.5 kN/m

References: model=BLD-EXTENSION; drawings=S-201; reports=R-201; REQ=REQ-103

Status: SIMULATION_ONLY / PROFESSIONAL ACCEPTANCE CRITERION NOT APPLIED

CAL-S06 ? Representative simple-span moment

Method: $\text{line_load} * \text{span}^2 / 8$

Result: 76.56 kNm

References: model=BLD-EXTENSION; drawings=S-201; reports=R-201; REQ=REQ-103

Status: SIMULATION_ONLY / PROFESSIONAL ACCEPTANCE CRITERION NOT APPLIED

CAL-S07 ? Representative support reaction

Method: $\text{line_load} * \text{span} / 2$

Result: 61.25 kN

References: model=BLD-EXTENSION; drawings=S-201; reports=R-201; REQ=REQ-103

Status: SIMULATION_ONLY / PROFESSIONAL ACCEPTANCE CRITERION NOT APPLIED

FOUNDATION

CAL-F01 ? Assumed bearing area

Method: $\text{strip_width} * \text{total_strip_length}$

Result: 72.0 m²

References: model=BLD-EXTENSION; drawings=S-101; reports=R-201; REQ=REQ-104

Status: SIMULATION_ONLY / PROFESSIONAL ACCEPTANCE CRITERION NOT APPLIED

CAL-F02 ? Indicative average contact pressure

Method: $\text{total_service_load} / \text{bearing_area}$

Result: 18.54 kPa

References: model=BLD-EXTENSION; drawings=S-101; reports=R-201; REQ=REQ-104

Status: SIMULATION_ONLY / PROFESSIONAL ACCEPTANCE CRITERION NOT APPLIED

EGRESS

CAL-E01 ? Total simulated exit width

Method: $\text{exit_count} * \text{exit_width_each}$

Result: 2.4 m

References: model=BLD-EXTENSION; drawings=F-101; reports=R-301; REQ=REQ-105

Status: SIMULATION_ONLY / PROFESSIONAL ACCEPTANCE CRITERION NOT APPLIED

CAL-E02 ? Persons per simulated exit

Method: $\text{peak_occupancy} / \text{exit_count}$

Result: 100.0 persons/exit

References: model=BLD-EXTENSION; drawings=F-101; reports=R-301; REQ=REQ-105,REQ-107

Status: SIMULATION_ONLY / PROFESSIONAL ACCEPTANCE CRITERION NOT APPLIED

CAL-E03 ? Persons per metre total exit width

Method: $\text{peak_occupancy} / \text{total_exit_width}$

Result: 83.33 persons/m

References: model=BLD-EXTENSION; drawings=F-101; reports=R-301; REQ=REQ-105,REQ-107

Status: SIMULATION_ONLY / PROFESSIONAL ACCEPTANCE CRITERION NOT APPLIED

VENTILATION

CAL-V01 ? Ventilation flow regular_future

Method: $\text{persons} * \text{rate_l_s_person} * 3.6$

Result: 3780.0 m³/h

References: model=BLD-EXISTING,BLD-EXTENSION; drawings=A-101,A-102; reports=R-301; REQ=REQ-105,REQ-107

Status: SIMULATION_ONLY / PROFESSIONAL ACCEPTANCE CRITERION NOT APPLIED

CAL-V02 ? Ventilation flow friday_future

Method: $\text{persons} * \text{rate_l_s_person} * 3.6$

Result: 3150.0 m³/h

References: model=BLD-EXISTING,BLD-EXTENSION; drawings=A-101,A-102; reports=R-301; REQ=REQ-105,REQ-107
Status: SIMULATION_ONLY / PROFESSIONAL ACCEPTANCE CRITERION NOT APPLIED

CAL-V03 ? Ventilation flow special_peak

Method: persons * rate_l_s_person * 3.6

Result: 5040.0 m3/h

References: model=BLD-EXISTING,BLD-EXTENSION; drawings=A-101,A-102; reports=R-301; REQ=REQ-105,REQ-107
Status: SIMULATION_ONLY / PROFESSIONAL ACCEPTANCE CRITERION NOT APPLIED

PARKING

CAL-P01 ? Available spaces regular_weekday

Method: capacity - occupied

Result: 115.0 spaces

References: model=P-A,P-B,P-C,P-D,P-E; drawings=A-001; reports=R-401; REQ=REQ-106
Status: SIMULATION_ONLY / PROFESSIONAL ACCEPTANCE CRITERION NOT APPLIED

CAL-P02 ? Available spaces friday_pre_peak

Method: capacity - occupied

Result: 90.0 spaces

References: model=P-A,P-B,P-C,P-D,P-E; drawings=A-001; reports=R-401; REQ=REQ-106
Status: SIMULATION_ONLY / PROFESSIONAL ACCEPTANCE CRITERION NOT APPLIED

CAL-P03 ? Available spaces friday_prayer_peak

Method: capacity - occupied

Result: 55.0 spaces

References: model=P-A,P-B,P-C,P-D,P-E; drawings=A-001; reports=R-401; REQ=REQ-106
Status: SIMULATION_ONLY / PROFESSIONAL ACCEPTANCE CRITERION NOT APPLIED

CAL-P04 ? Available spaces saturday_active_use

Method: capacity - occupied

Result: 80.0 spaces

References: model=P-A,P-B,P-C,P-D,P-E; drawings=A-001; reports=R-401; REQ=REQ-106
Status: SIMULATION_ONLY / PROFESSIONAL ACCEPTANCE CRITERION NOT APPLIED

CAL-P05 ? Available spaces sunday_active_use

Method: capacity - occupied

Result: 100.0 spaces

References: model=P-A,P-B,P-C,P-D,P-E; drawings=A-001; reports=R-401; REQ=REQ-106
Status: SIMULATION_ONLY / PROFESSIONAL ACCEPTANCE CRITERION NOT APPLIED

CAL-P06 ? Synthetic parking surplus regular_future

Method: minimum_available - demand

Result: 25.0 spaces

References: model=P-A,P-B,P-C,P-D,P-E; drawings=A-001; reports=R-401; REQ=REQ-106
Status: SIMULATION_ONLY / PROFESSIONAL ACCEPTANCE CRITERION NOT APPLIED

CAL-P07 ? Synthetic parking surplus friday_future

Method: minimum_available - demand

Result: 30.0 spaces

References: model=P-A,P-B,P-C,P-D,P-E; drawings=A-001; reports=R-401; REQ=REQ-106

Status: SIMULATION_ONLY / PROFESSIONAL ACCEPTANCE CRITERION NOT APPLIED

CAL-P08 ? Synthetic parking surplus special_peak

Method: minimum_available - demand

Result: 15.0 spaces

References: model=P-A,P-B,P-C,P-D,P-E; drawings=A-001; reports=R-401; REQ=REQ-106

Status: SIMULATION_ONLY / PROFESSIONAL ACCEPTANCE CRITERION NOT APPLIED

CONSTRUCTION ACTIVITY

CAL-C01 ? Construction-phase duration

Method: sum(duration_days excluding operational use)

Result: 95.0 days

References: model=BLD-EXTENSION; drawings=X-101; reports=R-501; REQ=REQ-108

Status: SIMULATION_ONLY / PROFESSIONAL ACCEPTANCE CRITERION NOT APPLIED

CAL-C02 ? Synthetic equipment operating hours

Method: sum(equipment operating_hours)

Result: 496.0 h

References: model=BLD-EXTENSION; drawings=X-101; reports=R-501; REQ=REQ-108

Status: SIMULATION_ONLY / PROFESSIONAL ACCEPTANCE CRITERION NOT APPLIED

CAL-C03 ? Synthetic transport activity

Method: sum(movements * distance_km)

Result: 6600.0 vehicle-km

References: model=BLD-EXTENSION; drawings=X-101; reports=R-501; REQ=REQ-108

Status: SIMULATION_ONLY / PROFESSIONAL ACCEPTANCE CRITERION NOT APPLIED

LIMITATIONS

LIM-01 ? CONCEPT CALCULATIONS - NOT FOR SUBMISSION OR EXECUTION

LIM-02 ? Geometry is a concept model and requires REQ-102 validation.

LIM-03 ? Loads, spans and structural system are synthetic test fixtures.

LIM-04 ? Contact pressure is indicative; resistance and settlement are not verified.

LIM-05 ? No Bbl or fire-safety compliance conclusion is made.

LIM-06 ? 225 spaces are project-leader-confirmed but field verification is pending.

LIM-07 ? Activity values are synthetic and no AERIUS deposition result is generated.